Multiple Linear Regression

Example: Restaurant prices

Consider an example studying the relationship between the average price of dinner at a restaurant, and the restaurant's food quality and service quality.

y: dinner price

x_1: food quality score, on 0-to-30 scale

x_2: service quality score, on 0-to-30 scale

The "restaurants" data set contains this data for 168 Italian restaurants in NYC in 2001.

Example: Restaurant prices

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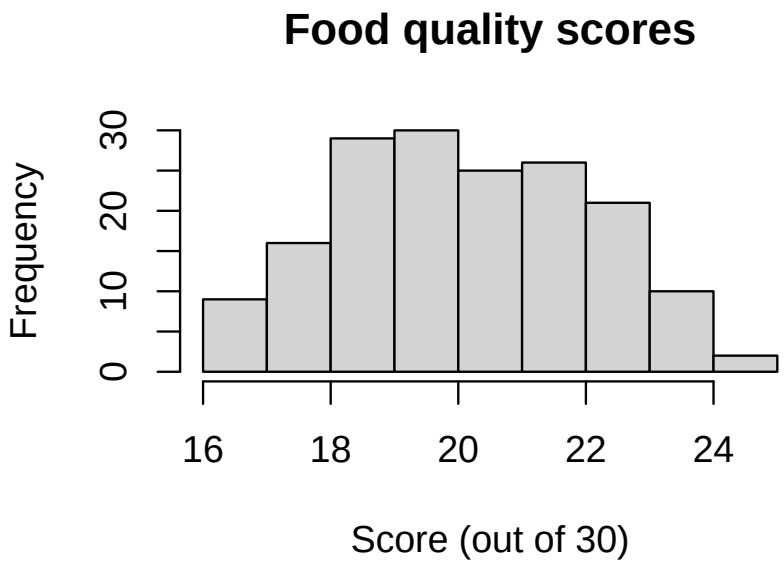
```
##   Price Food Service
## 1    43   22      20
## 2    32   20      19
## 3    34   21      18
## 4    41   20      17
## 5    54   24      21
## 6    52   22      21
```

Explore the data



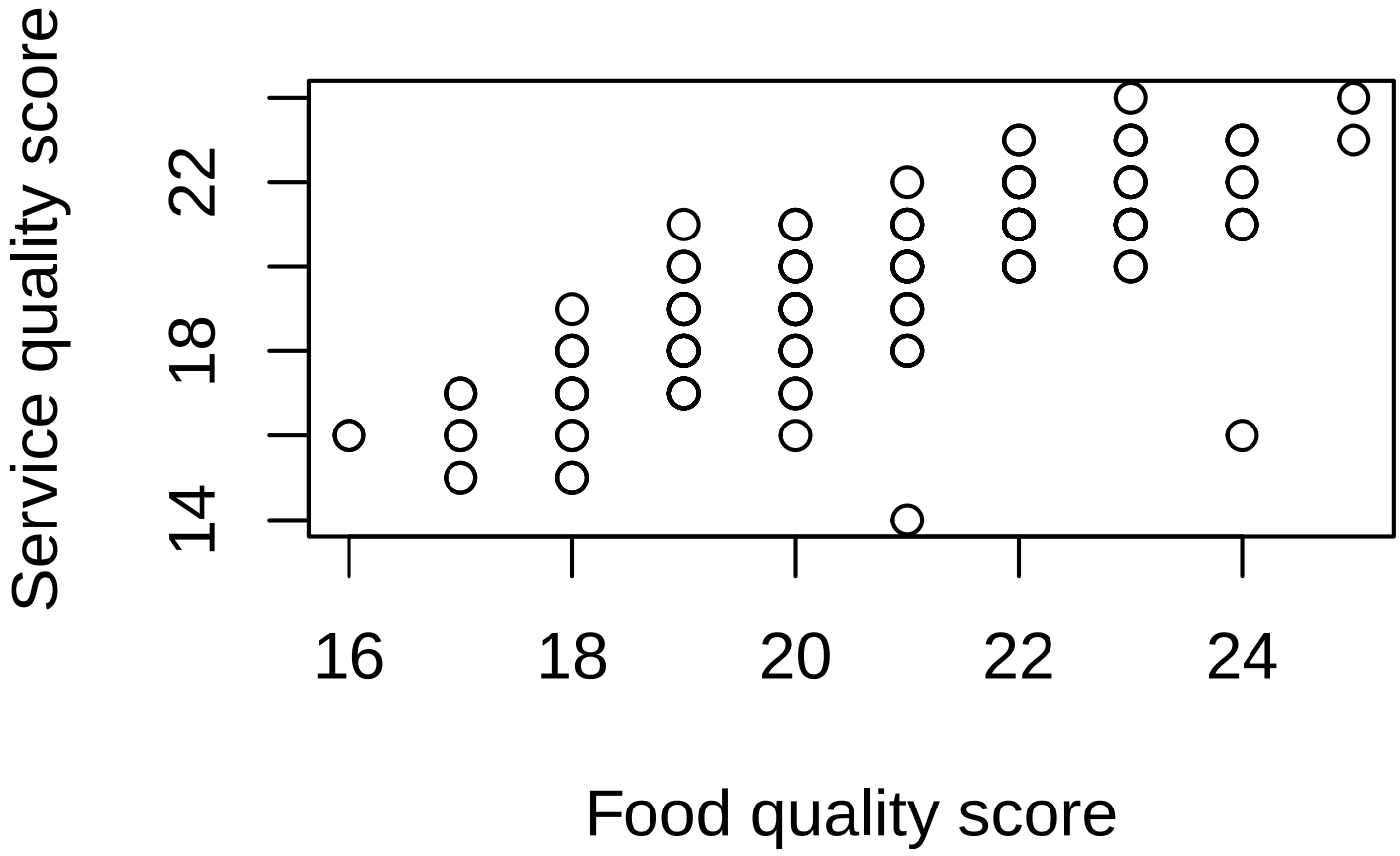
The least expensive restaurant had a dinner price of \$19, the most expensive was \$65.

Explore the data





Explore the data



The scores for food quality and service quality are positively correlated (not surprising).

Simple linear regression model

We could model Price using only food score as a predictor:

```
foodmodel<-lm(Price ~ Food, data=restaurants)
round(summary(foodmodel)$coefficients, 3)
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -17.832      5.863  -3.041    0.003
## Food           2.939      0.283  10.371    0.000
```

Based on this simple linear regression model, we estimate that restaurants with a one-point difference in food quality score have a dinner price difference of $ 2.94, on average.

Simple linear regression model

Or, we could model Price using only service score as a predictor:

```
servicemodel<-lm(Price ~ Service, data=restaurants)
round(summary(servicemodel)$coefficients, 3)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -11.978      5.109  -2.344     0.02
## Service        2.818      0.262  10.764     0.00
```

Based on this simple linear regression model, we estimate that restaurants with a one-point difference in service quality score have a dinner price difference of $ 2.82, on average.

Multiple linear regression model

What if we construct a single model using both food quality and service as predictors?

To specify multiple predictors in the regression model, we include them by name in the lm() function, separated by a "+" symbol.

```
combined_model<-lm(Price ~ Food + Service,
                   data=restaurants)
```

Multiple linear regression: estimated model

```
round(summary(combined_model)$coefficients, 3)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -21.159      5.665  -3.735    0.000
## Food           1.495      0.446   3.351    0.001
## Service        1.704      0.418   4.072    0.000
```

In the combined model, we estimate that for restaurants with the same service quality and a one-point difference in food quality, on average dinner costs \$1.50 more at the restaurant with the better food.

We estimate that for restaurants with the same food quality and a one-point difference in service quality, on average dinner costs \$1.70 more at the restaurant with the better service.

Multiple linear regression: estimated model

```
round(summary(combined_model)$coefficients, 3)
```

```
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -21.159      5.665  -3.735    0.000
## Food           1.495      0.446   3.351    0.001
## Service        1.704      0.418   4.072    0.000
```

Because the food score and the service score are on the same scale, it makes sense to say that there is a bigger difference in price associated with service quality, compared to food quality.

Alternate R syntax for multiple regression

When we want to fit a multiple linear regression model to a data set in R, and we want to use all of the variables in the data frame as predictors (except the response variable), it can be inconvenient to type out all of their names.

There is another option:

```
alt_model<-lm(Price~., data=restaurants)
```

The shorthand "y ~." tells the lm( ) function to use all of the variables in the data frame EXCEPT y as predictors.

Alternate R syntax for multiple regression

We can confirm that this produces the same model:

```
round(summary(alt_model)$coefficients, 3)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -21.159      5.665  -3.735    0.000
## Food           1.495      0.446   3.351    0.001
## Service        1.704      0.418   4.072    0.000
```

Inference in multiple regression models

```
round(summary(combined_model)$coefficients,3)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -21.159      5.665  -3.735    0.000
## Food           1.495      0.446   3.351    0.001
## Service        1.704      0.418   4.072    0.000
```

The p values in the summary table correspond to individual hypothesis tests:

$H_0 : \beta_k = 0$ vs $H_a : \beta_k \neq 0$,

for each of the coefficients individually.

Making predictions

We can use the fitted model to predict the typical dinner price for a restaurant with food score 28 and service score 28:

```
predict(combined_model,
        newdata=data.frame('Food'=28, 'Service'=28))
```

```
##        1
## 68.42658
```

How does this compare to the predictions from the individual models?

```
predict(foodmodel, newdata=data.frame('Food'=28))
```

```
##        1
## 64.45873
```

```
predict(servicemodel, newdata=data.frame( 'Service'=28))
```

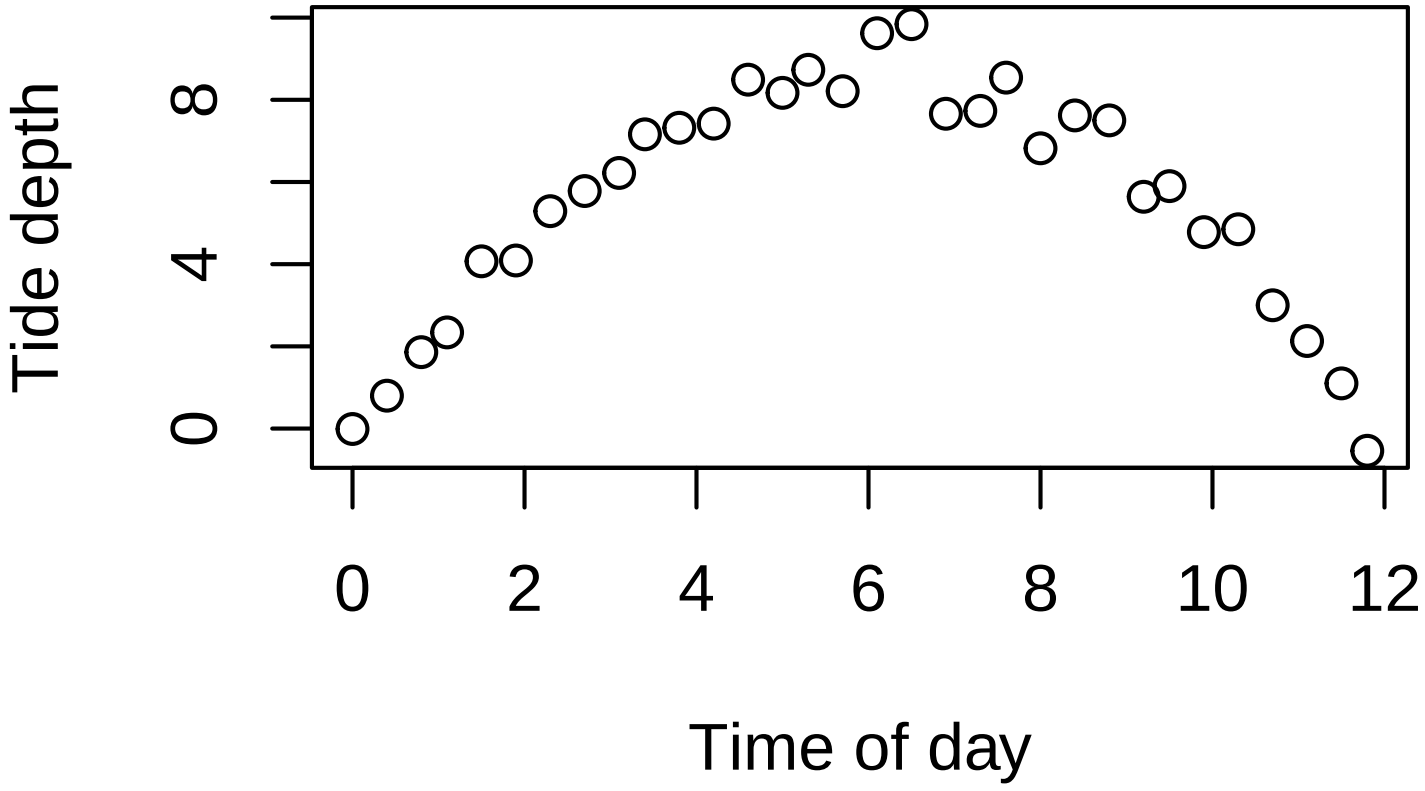
```
##        1
## 66.93831
```

Making predictions

In the combined model, we predict a slightly higher dinner price ($68.43) for a restaurant with excellent food *and* service, compared to the predictions we would make based on the food quality alone ($64.46) or the service quality alone ($66.94).

Example: transformations

Consider the following data consisting of tide depth measurements taken at a coastal location at different times throughout the day:



The relationship between time of day and tide depth is non-linear.

Example: transformations

What if we include time^2 as a new predictor in the model?

$$\text{depth} = \beta_0 + \beta_1 * \text{time} + \beta_2 * \text{time}^2 + \varepsilon$$

Example: transformations

To include x^2 as a predictor, the input to the lm( ) function looks a bit different:

```
tidemodel<-lm(depth ~ time + I(time^2), data=tides)
```

This is a quirk: exponents have a different meaning inside the lm( ) function, so to specify that we want time^2, we "isolate" it by wrapping it inside the I( ) expression.

Example: transformations

```
round(summary(tidemodel)$coefficients,3)
```

```
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)      -0.420      0.262  -1.601     0.12
## time              3.049      0.103  29.743     0.00
## I(time^2)        -0.254      0.008 -30.372     0.00
```

```
summary(tidemodel)$r.squared
```

```
## [1] 0.9695942
```

With the new predictor, the model now has $R^2 = 0.9696$, a much better fit compared to what we would get with a simple linear regression model.

Example: transformations

